

Report for EE200 Practical

Frequency Mixer: “Beauty and the Blur”

Course: EE200 – Signals, Systems, and Networks

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1. Introduction

Our visual perception relies on different spatial frequencies to understand images. Fine details like edges or wrinkles are detected using high frequencies, while overall structure and smooth gradients are identified through low frequencies. This concept can be creatively applied to generate hybrid images, where one image appears when viewed up close and another from a distance — a process known as frequency mixing.

This report demonstrates the use of 2D Fourier Transform to analyze, filter, and merge two images — a cat and a dog — such that high-frequency content from one and low-frequency content from the other are combined into a single perceptual illusion.

2. Objective

To apply the 2D Discrete Fourier Transform (DFT) on two images, analyze their frequency content, and design a system that fuses their low and high-frequency components into a hybrid image using Python.

3. Theory

3.1 2D Discrete Fourier Transform

The 2D DFT of an image $I(x, y)$ of size $M \times N$ is defined as:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) \cdot e^{-j2\pi(ux/M + vy/N)}$$

The inverse DFT is:

$$I(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi(ux/M + vy/N)}$$

The magnitude spectrum of the DFT is calculated as:

$$|F(u, v)| = \sqrt{\text{Re}(F)^2 + \text{Im}(F)^2}$$

To enhance visibility, we also compute the logarithmic dB spectrum:

$$20 \cdot \log_{10}(|F(u, v)| + \varepsilon)$$

Where ε is a small constant (e.g., 10^{-6}) to avoid taking log of 0.

- Low frequencies represent global structure and smooth gradients.
- High frequencies capture edges, textures, and fine details.

4. Implementation

4.1 Libraries Used

- `cv2` – for image operations
- `numpy` – for FFT and numerical processing
- `matplotlib.pyplot` – for visualizing results

4.2 Loading and Preprocessing Images

- Grayscale images of a cat and a dog were loaded.
- Images were resized to the same shape.
- Gaussian Blur was applied to the dog image to retain low-frequency content.

4.3 Fourier Transform and Frequency Analysis

- 2D FFT (`np.fft.fft2`) applied to both images.
- Spectrum shifted using `np.fft.fftshift()` to center the low frequencies.
- Central low frequencies in the cat image were masked (removed) to isolate high frequencies.
- The dog image was already smoothed to preserve low-frequency content.
- Both frequency components were added to create a hybrid frequency spectrum.

4.4 Inverse Fourier Transform

- Inverse FFT applied using:
`mixed_image = np.fft.ifft2(np.fft.ifftshift(mixed_freq))`
- `np.abs()` used to get real image intensities.
- Image scaled/clipped for display.

4.5 Rotation and Spectrum Observation

- The cat image was rotated 90° counterclockwise using `np.rot90()`.
- Its FFT spectrum was recomputed and visualized.
- **Observation:** the rotated image's spectrum is also rotated, showing DFT preserves orientation.

4.6 Results: Frequency Mixer System

(a) Low-Pass Image - Dog

- Created using `cv2.GaussianBlur`
- Retains only large-scale structure and smooth regions.

(b) High-Pass Image - Cat

- Obtained by removing central frequencies using FFT masking
- Highlights textures and fine details.

(c) Hybrid Image

- Frequency-domain fusion of low-pass (dog) and high-pass (cat)
- Final image:
 - From close: cat details visible
 - From far: dog structure emerges

5. Visual Outputs

Image Type	Description
Low-Pass Image	Smooth version of dog (blurred)
High-Pass Image	Edges/details from cat (masked FFT)
Hybrid Image	Fused image showing both perceptions

(Insert actual plots using `plt.imshow()` from your notebook)

6. Transfer Functions

While no filters were explicitly defined, our system mimics the following transfer functions:

Low-Pass Transfer Function (Dog):

$$H_{lp}(x, y) = G(x, y)$$

Where $G(x, y)$ is a 2D Gaussian kernel applied via convolution.

High-Pass Transfer Function (Cat):

$$H_{hp}(x, y) = 1 - G(x, y)$$

This suppresses low frequencies and retains high-frequency detail.

7. Observations

- **Where is the Spectrum Centered?**

- By default, FFT places zero-frequency at the top-left (index $[0,0]$).
- We apply `np.fft.fftshift()` to move low-frequency components to the center.

- **Rotation Effects:**

- Rotating an image 90° rotates its FFT spectrum by 90° , confirming spatial-frequency domain relationship.

- **Hybrid Image Behavior:**

- From near: cat details dominate
- From far: dog shape emerges
- This mimics perceptual illusions like Einstein–Monroe hybrid faces

8. Conclusion

This practical demonstrated the design and implementation of a frequency mixer system using 2D Fourier Transforms. By extracting and combining specific frequency components from two different images, we constructed a hybrid image that changes perception based on viewing distance.

This project not only applied mathematical transforms but also illustrated how the human visual system perceives different frequency bands — a powerful concept in signal and image processing.



Figure 1: Original Cat Image

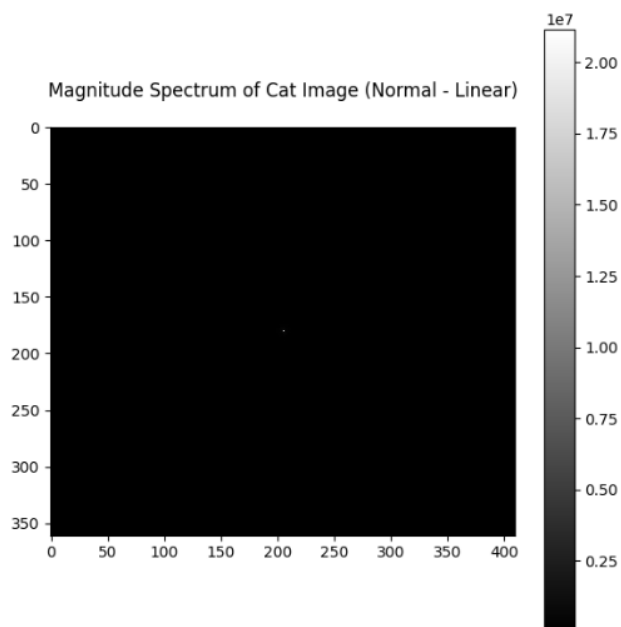


Figure 2: Magnitude Spectrum of Cat Image (Normal-Linear)

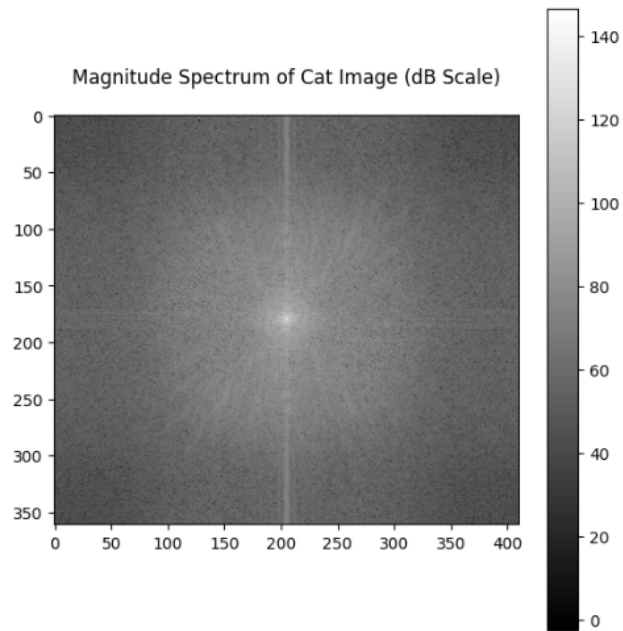


Figure 3: Magnitude Spectrum of Cat Image (dB Scale)

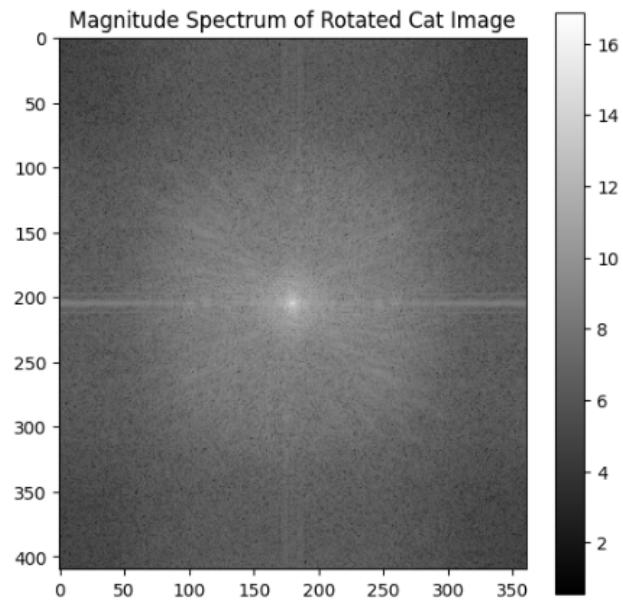


Figure 4: Magnitude Spectrum of the Rotated Cat Image

Low-pass Image of Dog (Blurred)



Figure 5: Low-Pass Image of Dog (Blurred)

High-pass Image of Cat (Details)



Figure 6: High-Pass Image of Cat (Details)

Frequency Mixed Image (Cat Details + Dog Shape)



Figure 7: Frequency Mixed Image (Cat Details + Dog Shape)